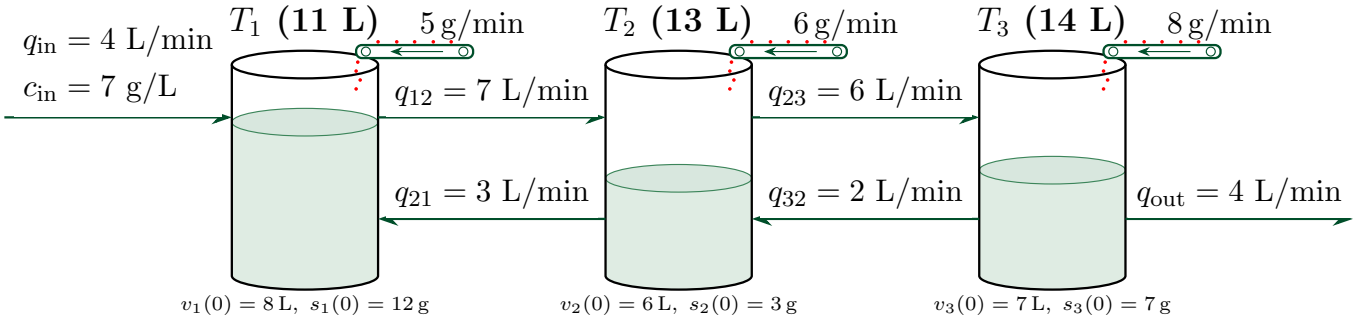
1

MA3521 — Exam One

Name:



Tanks are well-mixed, t in min, $\mathbf{v}(t)$ in L, $\mathbf{s}(t)$ in g, and conveyers add dry salt at the rate shown.

(a) $\mathbf{s}' = \boldsymbol{A}\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \begin{pmatrix} 12 \\ 3 \\ 7 \end{pmatrix}$ g $\quad 0 < t <$ [ ] min

$\mathbf{v}(t) =$ [ ] L $\qquad \mathbf{b} =$ [ ] g/min

(b) $\boldsymbol{A} =$ [ ] min^{-1}

(c) A UC form is $\mathbf{s}_p(t) =$ [ ]. Constants are determined by row reducing $\boldsymbol{M}_{\text{UC}} =$ [ ]

(a) ☐ Linear ☐ Separable $\quad ty' - 3y = t^4 e^{2t} \quad y(1) = 1$

Show a suitable intermediate step

Show a fully integrated answer

(b) ☐ Linear ☐ Separable $\quad y' = 3t^2 \cos(t^3)(1+y)^2 \quad y(0) = 1$

Show a suitable intermediate step

Show a fully integrated answer

3

$\mathbf{y}' = \boldsymbol{A}\mathbf{y} + e^{3t}\begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix} 1 & -2 \\ 8 & 1 \end{pmatrix} \quad \mathbf{y}(0) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

Eigenpairs $1 + 4i$, $\begin{pmatrix} 1 \\ -2i \end{pmatrix}$ $\quad 1 - 4i$, $\begin{pmatrix} 1 \\ 2i \end{pmatrix}$

(a) $\mathbf{y}_c(t) =$

(b) A suitable UC form is

$\mathbf{y}_p(t) =$

(c) To determine the UC coefficients you would row reduce

$\boldsymbol{M}_{\text{UC}} =$

(d) To satisfy the IC you would row reduce

$\boldsymbol{M}_{\text{IC}} =$

$\mathbf{y}' = \boldsymbol{A}\mathbf{y} + \sin(3t)\begin{pmatrix}2\\1\end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix}2 & 1\\0 & -1\end{pmatrix}$

Eigenpairs 2, $\begin{pmatrix}1\\0\end{pmatrix}$ $\quad -1, \begin{pmatrix}1\\-3\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$

(b) A suitable UC form is

$\mathbf{y}_p(t) =$

(c) To determine the UC coefficients you would row reduce

$\boldsymbol{M}_{\text{UC}} =$